

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): Model:  $\frac{dN}{dt} = rN(1 - \frac{N}{K})$  N: population at time t; K: carrying capacity (depends on resources);  $r, K > 0$ .

Sol:  $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \rightarrow K$  Equilibrium points:  $N = 0$  (unstable),  $N = K$  (stable). Quickest:  $N = \frac{K}{2}$  (<凹; >凸)

Delay Model: Model:  $\frac{dN}{dt} = f(N(t), N(t - T))$  T: delay time > 0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

1 Ext. of Logistic Model:  $\frac{dN}{dt} = rN(t) [1 - \frac{N(t-T)}{K}]$  (用"ave. pop." 估计, 准用"Convolution Type")  $\Rightarrow$  2 Dimensionless:  $\frac{dN^*}{dt^*} = N^*(t^*) [1 - N^*(t^* - T^*)]$  ( $N^* = \frac{N}{K}$ ,  $t^* = rt$ ,  $T^* = rT$ )

1,2 Periodic sol: stable limit cycle  $N(t + t_p) = N(t)$ ,  $t_p$ : period. Amplitude:  $rT$  Equilibrium Points:  $N = 0 | N^* = 0$  (unstable);  $N = K | N^* = 1$  (Stable:  $0 < T^* = rT < \frac{\pi}{2}$ ; Unstable:  $T^* > \frac{\pi}{2}$ )

Continuous Population Model (EN): Model:  $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$   $r, K, E > 0$ ; EN: harvesting rate/unit time; E: effort measure.

Equilibrium:  $N = 0$  (unstable);  $N = N_h(E) = K(1 - \frac{E}{r}) > 0$  (need  $E < r$ , Stable) Yield:  $Y(E) = EN \cdot E > r$ : die out; else: Max Yield:  $Y_M = \frac{rK}{4}$  ( $E = \frac{r}{2}$ ) Steady State:  $N_h | Y_M = \frac{K}{2}$

Continuous Population Model (Y0): Model:  $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$   $Y_0$ : constant yield rate/unit time.

Max Yield:  $Y_M = \frac{rK}{4}$  Equilibrium:  $0 < Y_0 < Y_M$ : Two positive points (unstable(小) and stable(大)). Sensitivity: 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of  $N_{t+1} = f(N_t)$ : Verhulst:  $N_{t+1} = rN_t(1 - \frac{N_t}{K})$ ;  $r, K > 0$ . Better: Ricker curve:  $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

Imperfect Bifurcation:  $\dot{x} = rx - x^3 + h$  (codimension-2 bifurcation) If  $r < 0$ ,  $x^*$  unique. If  $r > 0$ ,  $|h| > h_c$ ,  $x^*$  unique;  $|h| < h_c$ ,  $x^*$  three. (两边 stable, 中 unstable). Bifurcation points:  $h = \pm h_c = \pm \frac{2r}{3} \sqrt{\frac{r}{3}}$ . cusp point:  $(r, h) = (0, 0)$

1.2 Summary - Stability and Bifurcation

|                                                   |                                                                                                                                                                                                |                                                                          |                                                                              |                                                                                  |                                                                                                                                                            |  |                                           |  |                                                                                                                              |  |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------|--|
|                                                   | $\frac{dN}{dt} = f(N)$ <b>No Limit cycle periodic solutions</b> $S$ : stable; $U$ : unstable.                                                                                                  |                                                                          |                                                                              |                                                                                  | $N_{t+1} = N_t F(N_t) = f(N_t)$ <b>Eigenvalue:</b> $\lambda = f'(N^*)$                                                                                     |  |                                           |  |                                                                                                                              |  |
| Stability                                         | Stable: $f'(N^*) < 0$                                                                                                                                                                          |                                                                          | Unstable: $f'(N^*) > 0$                                                      |                                                                                  | Stable: $ \lambda  < 1$                                                                                                                                    |  | Unstable: $ \lambda  > 1$ (chaotic)       |  | Periodic: $ \lambda  = 1 \begin{cases} + : \text{tangent bifurcation} \\ - : \text{period-doubling bifurcation} \end{cases}$ |  |
| Linearise                                         | $N(t) = N^* + n(t),  n(t)  \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = \begin{cases} \text{(general)} = f(N^*) + f'(N^*)n(t) \\ \text{(delay)} = \text{直接带入 } t \text{ 后近似} \end{cases}$ |                                                                          |                                                                              |                                                                                  | Let $N_t = N^* + n_t,  n_t  \ll 1 \Rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \Rightarrow n_{t+1} = \lambda n_t$          |  |                                           |  |                                                                                                                              |  |
| Bifurcation \\\ Perturbations at $N^*$   Periodic | Saddle-Note (Normal Form)<br>$\dot{x} = r \pm x^2$<br>$0 \rightarrow S \rightarrow U$                                                                                                          | Transcritical<br>$\dot{x} = rx - x^2$<br>$S \rightarrow U \rightarrow S$ | (super-)Pitchfork<br>$\dot{x} = rx - x^3$<br>$S \rightarrow S \rightarrow U$ | Sub-critical Pitchfork<br>$\dot{x} = r + x^3$<br>$U \rightarrow U \rightarrow S$ | $\lambda < -1$ : $\uparrow$ oscillations $-1 < \lambda < 0$ : $\downarrow$ oscillations                                                                    |  | $0 < \lambda < 1$ : $\downarrow$ Monotone |  | $\lambda > 1$ : $\uparrow$ Monotone                                                                                          |  |
|                                                   |                                                                                                                                                                                                |                                                                          |                                                                              |                                                                                  | If $fP(N^*) = N^*$ , $N^*$ is $p$ -periodic. $\vee$ even $p$ -periodic $N^*$ with $\lambda = -1$ branches into $2p$ -periodic                              |  |                                           |  |                                                                                                                              |  |
| Other                                             | <b>Time-Independent:</b> At $N^*$ : $\frac{dN}{dt} = 0$ $N(t) = N(t-T) = N^*$ $f(N^*) = 0$                                                                                                     |                                                                          |                                                                              |                                                                                  | <b>Sarkovskii's Theorem:</b> If a 3-periodic $N^*$ exists, then $\forall p \geq 1$ , $p$ -periodic $N^*$ exists. <b>Species extinction:</b> $N_{\min} < 1$ |  |                                           |  |                                                                                                                              |  |
| Other                                             | <b>Tangency Condition:</b> If $\dot{x} = f(x, r)$ , 分岔点 $(x^*, r_c)$ : $\frac{\partial f}{\partial x}(x^*, r_c) = 0$ $f(x^*, r_c) = 0$      两点合并: <b>coalesce</b> .                            |                                                                          |                                                                              |                                                                                  | <b>Max/Min:</b> [如果单峰 (unimodal)] 2-periodic $f'(N_m) = 0$ , $N_{\max} = f(N_m)$ ; $N_{\min} = f(N_{\max})$                                                |  |                                           |  |                                                                                                                              |  |

Hopf Bifurcation: 平衡点从稳定变为不稳定, 同时产生 limit cycle. 条件:  $\det(J) > 0$ ;  $\frac{d\text{tr}(J)}{d\mu} \neq 0$  or:  $\text{tr}(J)$  +/- 符号有变化  $\Leftrightarrow \lambda = \pm i\omega$  and  $\frac{d\Re(\lambda)}{d\mu} \neq 0$   $\mu$ : parameter, not time.

$\Rightarrow$  Stable  $\rightarrow$  Unstable + Stable Limit Cycle (supercritical) or Unstable  $\rightarrow$  Stable + Unstable Limit Cycle (subcritical). (一般 super-) when  $\text{tr}(J) = 0 | \Re(\lambda) = 0$ ,  $\lambda = \pm i\omega \Rightarrow$  periodic:  $T = \frac{2\pi}{\omega}$

Bifurcation: Qualitative changes in dynamics are called bifurcations. (当平衡点性质 | 个数发生变化时) Bifurcation Diagram: y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 | For Continuous Model: Find perturbation:  $n(t)$  and  $\frac{dn}{dt} \Rightarrow$  Assume  $n(t) = e^{\lambda t} | ce^{\lambda t} \rightarrow$  Find  $\lambda = g(\lambda) \Rightarrow$  Let  $\lambda = u + i\omega \rightarrow u = g_1(u)$  and  $\omega = g_2(\omega) \Rightarrow u < 0$  asy.stable;  $u > 0$  unstable. Find  $T_c$ : (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check  $T = 0$  stable.  $\Rightarrow$  Using  $u = 0$  (i.e.  $\lambda = i\omega$ ) to find  $T_c$ .

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey):  $\frac{dN}{dt} = N(a - bP)$ ;  $\frac{dP}{dt} = P(cN - d) \Rightarrow$  nondim:  $\frac{dN}{d\tau} = u(1 - v)$ ;  $\frac{dP}{d\tau} = \alpha v(u - 1)$  Equilibrium:  $(u^*, v^*) = (0, 0), (1, 1)$

Exact sol:  $\alpha u + v - \ln(u^\alpha v) = H$ ,  $H > H_{\min} = \alpha + 1$  constant. (closed phase) Conservation Law: Using  $\frac{du}{dv} = \frac{du}{d\tau} \frac{d\tau}{dv}$  find  $H \Rightarrow \frac{dH}{d\tau} = 0$ , here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If  $H(u, v)$  is level curve, Look  $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix}_{(u^*, v^*)} \Rightarrow$  If +/- definite (i.e.  $\forall \lambda_i > 0 (< 0)$ )  $\Rightarrow$  It's local min/max at  $(u^*, v^*)$  & strictly convex/concave  $\Rightarrow$  closed curves. Closed curves  $\Rightarrow$  periodic sol.

Determinant Stability: For model:  $\begin{bmatrix} u'(t) \\ v'(t) \end{bmatrix} = \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix}$  at equilibrium  $(u^*, v^*)$ . Let  $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u - u^* \\ v - v^* \end{bmatrix}$ , then  $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix}_{(u^*, v^*)} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow$  Find eigenvalues  $\lambda$  of  $J$  by  $\det(J - \lambda I) = 0$

1  $\det(J) > 0$ ,  $\text{tr}(J) < 0 \Rightarrow \Re(\lambda) < 0 \rightarrow$  stable node  $\text{Im}(\lambda) = 0$ ; stable focus=spiral point  $\text{Im}(\lambda) \neq 0$ ; 2  $\det(J) > 0$ ,  $\text{tr}(J) > 0 \Rightarrow \Re(\lambda) > 0 \rightarrow$  unstable node  $\text{Im}(\lambda) = 0$ ; unstable focus=spiral point  $\text{Im}(\lambda) \neq 0$

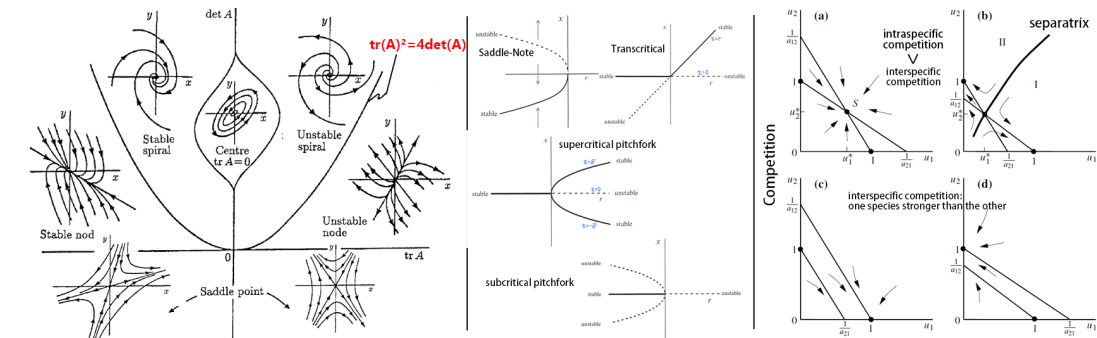
3  $\det(J) < 0 \Rightarrow \text{Im}(\lambda) = 0$ ,  $\lambda_1 < 0 < \lambda_2 \rightarrow$  saddle point (unstable); 4  $\det(J) > 0$ ,  $\text{tr}(J) = 0 \Rightarrow \lambda = \pm i\omega$  (pure imaginary)  $\rightarrow$  center (only linear stability) | Hopf bifurcation (见前面)

Nullclines:  $f(u, v) = 0$  or  $g(u, v) = 0 \rightarrow$  得 u,v 表达式是 nullclines. Trapping Region: A closed region R 可以用直线/null clines 构成 s.t. 所有 边界的外法线方向 n, 在对应边界上的点, 满足  $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

Find slope of Nullclines:  $f(u, v) = 0 \Rightarrow df = f_u du + f_v dv = 0 \Rightarrow \frac{dv}{du} = -\frac{f_u}{f_v}$  Similarly:  $\frac{dv}{du} = -\frac{g_u}{g_v}$

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in trapping region  $\Rightarrow$  A stable limit cycle exists. (Cannot be a saddle (since  $\det(J)$  cannot be  $< 0$ ))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (b,c,d)



3 Reaction Kinetics

Enzyme Kinetics: e.g.  $S + E \xrightleftharpoons[k_{-1}]{k_1} SE \xrightarrow{k_2} P + E$   $K_i$ : rate of reaction; S: substrate; E: enzyme; SE: complex; P: product. Rate of uptake:  $v = \frac{d[P]}{dt} |_{t=0}$

Law of Mass Action: Rate of reaction is proportional to the product of the concentrations of the reactants. e.g.  $\alpha_1 A + \alpha_2 B \xrightleftharpoons[k_{-1}]{k_1} \beta_1 C + \beta_2 D \Rightarrow$  rateforward  $= k_1 [A]^{\alpha_1} [B]^{\alpha_2}$ ; ratebackward  $= k_{-1} [C]^{\beta_1} [D]^{\beta_2}$

Quasi-Steady State Approximation (QSSA): 如果系统里有快和慢变量, 则可以假设快变量达到稳态, 即  $\frac{d(\text{fast})}{dt} = 0$ . (一般说来假设 complex concentration changes rapidly)

Singular Perturbations Analysis: For  $\frac{du}{d\tau} = f(u, v)$ ,  $\varepsilon \frac{dv}{d\tau} = g(u, v)$ ,  $0 < \varepsilon \ll 1 \Rightarrow v$  fast (when  $g \neq 0$ ),  $u$  slow. Quasi-steady-state approximation:  $\varepsilon \rightarrow 0 \Rightarrow g(u, v) = 0$

Taylor:  $u(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n u_n(\tau)$ ;  $v(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n v_n(\tau)$  Outer solution: Taylor 带入原方程组  $\rightarrow$  保留  $\mathcal{O}(1) | \varepsilon = 0$  Valid for  $\tau > 0$  (slow time);  $v_0(0)$  invalid.  $\rightarrow$  not uniformly valid for  $\tau \geq 0$

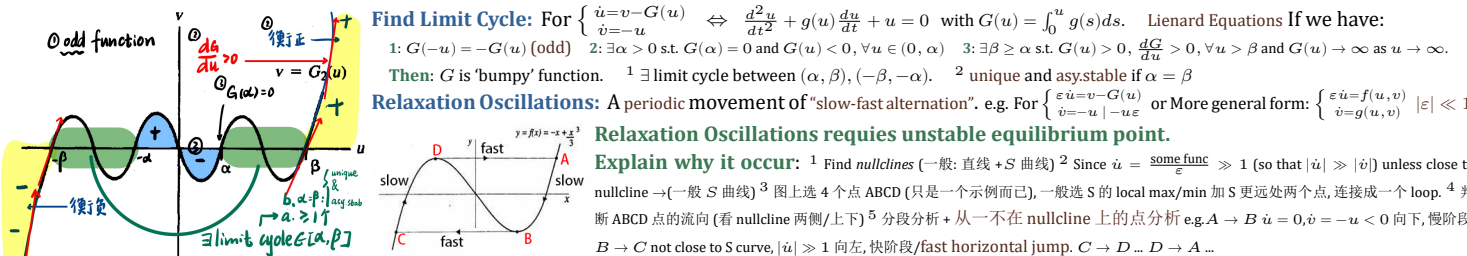
Inner solution:  $\sigma = \frac{\tau}{\varepsilon}$ ,  $V(\sigma; \varepsilon) = v(\tau; \varepsilon) \Rightarrow \frac{dV}{d\sigma} = \varepsilon f(u, v)$ ,  $\frac{dV}{d\sigma} = g(u, v) \rightarrow$  Taylor 带入原方程组  $\rightarrow$  保留  $\mathcal{O}(1) | \varepsilon = 0$  (known as boundary layer ( $\tau = 0$  处)); Valid  $\tau = 0$  (fast time)

Asymptotic Matching:  $\lim_{\sigma \rightarrow \infty} [U_0(\sigma), V_0(\sigma)] = \lim_{\tau \rightarrow 0^+} [u_0(\tau), v_0(\tau)]$  Uniformly Valid Solution:  $u(\tau; \varepsilon) = u_0(\tau) + \mathcal{O}(\varepsilon)$ ;  $v(\tau; \varepsilon) = \begin{cases} v_0(\tau) + \mathcal{O}(\varepsilon), & 0 < \varepsilon \ll \tau \\ V_0(\sigma) + \mathcal{O}(\varepsilon), & 0 < \tau \ll 1 \end{cases}$  uniform expansion: 见 chp.5

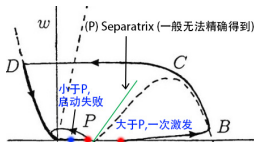
Activator-Inhibitor Model: For  $\frac{du}{dt} = f(u, v)$ ,  $\frac{dv}{dt} = g(u, v)$ : If  $\frac{\partial g}{\partial u} > 0$ ,  $u$  is an activator of  $v$ ; If  $\frac{\partial f}{\partial v} < 0$ ,  $v$  is an inhibitor of  $u$ . ps: Possible have bifurcation behaviour

Positive/Negative Feedback: At Equilibrium: Positive:  $f_u, f_v \geq 0$ ;  $g_u, g_v \leq 0$  or  $f_u, f_v \leq 0$ ;  $g_u, g_v \geq 0$  Negative:  $f_u, f_v \geq 0$ ;  $g_u, g_v \geq 0$  or  $f_u, f_v \leq 0$ ;  $g_u, g_v \leq 0$

4 Relaxation Oscillations & Switch Behaviour



Estimate period  $T$ : 例如上图 (例子忽略了快反应)  $T = \int_{t_A}^{t_B} dt + \int_{t_C}^{t_D} dt = 2 \int_{t_A}^{t_B} dt$  Find  $dt = _du$  带入计算 e.g. In slow time:  $\dot{u} = 0 \rightarrow v = G(u)$  and  $\dot{v} = -u \rightarrow \frac{dv}{dt} = \frac{dv}{du} \frac{du}{dt} = G'(u) \frac{du}{dt} = -u \rightarrow -\frac{G'(u)}{u} du = dt$ , 带入计算



**Excitable System: FitzHugh-Nagumo Model**  $\begin{cases} \frac{dx}{dt} = c[y + x - \frac{x^3}{3} + z(t)] \\ \frac{dy}{dt} = \frac{-x + a + by}{c} \end{cases}$  ( $x$ : excitability variable;  $y$ : recovery variable;  $z(t)$ : external stimulus)

**Threshold 现象:** **Small perturbation:** return to equilibrium (failed initiation/excitation). **Large perturbation:** large excursion before returning to equili. (action potential|successful initiation). After excitation| 大扰动之后: Refractory Period 不应期: system is unresponsive to further stimulation. **constant stimulus:** periodic firing (limit cycle).

**Reason:** 寻找  $\dot{u}, \dot{v}$  s.t.  $\ll, \gg 1$  i.e.  $|\dot{u}| \gg |\dot{v}|$  or  $|\dot{u}| \ll |\dot{v}|$

**Switch Behaviour:** 从一个 stable point 到另一个 stable point. (前面 excitable system|relaxation oscillation 如有两 stable points 且画图分析 (像 relaxation oscillation 那样分析) 可得到)

**描述话语 example:** 快-慢震荡: <sup>1</sup> (option) The equilibrium point  $S$  is unstable between the two knees/turning points of the  $u$ -nullcline/cubic-like curve. | or:  $S$  changes stability as it crosses the knees of the  $u$ -nullcline. <sup>2</sup> 图上标/简单解释拐点位置. <sup>3</sup> 证明  $|u'| \gg 1$  (写成乘积形式判断) 或其他  $\rightarrow$  得快运动边. <sup>4</sup> Trajectories starting off the  $u$ -nullcline rapidly/quickly approach left/right branch of the  $u$ -nullcline, and then slowly move down/up|crawl along the  $u$ -nullcline due to  $v' < 0 / > 0$ , and to upper/lower knee/turning point <sup>5</sup> At the knee/turnning point, the trajectory quickly jumps to the right/left branch of the  $u$ -nullcline, and then slowly moves up/down|crawl along the  $u$ -nullcline due to  $v' > 0 / < 0$ . <sup>6</sup> And this process repeats periodically, giving rise to a relaxation oscillation. **Excitable:** 类似的, 分析 small perturbation above/below equilibrium point  $S \rightarrow$  得 failed excitation or successful excitation.

## 5 Manifolds & Fenichel's Theorem | SIR Model

**Regular/Singular Perturbation:** 对于一个含  $\varepsilon$  的方程/系统: If  $\varepsilon \rightarrow 0$  the order does not change, it's a **regular perturbation**. If  $\varepsilon \rightarrow 0$  the order changes, it's a **singular perturbation**.

**Find Roots:** Assume roots  $x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$ , 代入比较  $\mathcal{O}(1), \mathcal{O}(\varepsilon) \dots$  得系数(解前两系数就行). If singular, cannot approx  $x = \mathcal{O}(\varepsilon^{-1})$ , rescaling  $x = \frac{X}{\varepsilon}$ , assume root  $X = X_0 + \varepsilon X_1 + \dots$  类似前. 不留  $X_0 = 0$  解, 因为  $X$  小.

**Sol & Uniform:** 对  $\varepsilon \dot{y} + p(x)\dot{y} + q(x)y = r(x)$ , a. 类似上面算 sol (有初值) or b. 直接算 exact sol.  $\rightarrow$  得 **Outer Solution** (首项含  $\varepsilon$ ). Let  $\tau = \varepsilon t$  类似算 sol  $\rightarrow$  **Inner Solution** (首项不含  $\varepsilon$ ). **unif expr** = Outer + Inner - Matching(=matching cond the lim)

**Singular Perturbation System:** For  $\begin{cases} \dot{u} = f(u, v, \varepsilon) \\ \dot{v} = g(u, v, \varepsilon) \end{cases}, 0 < \varepsilon \ll 1. \text{ If } \tau = \varepsilon t, \text{ then } \begin{cases} \varepsilon \dot{u}' = f(u, v, \varepsilon) \\ \dot{v}' = g(u, v, \varepsilon) \end{cases} \quad \begin{matrix} t: \text{fast time}; \tau: \text{slow time} \\ (\varepsilon \neq 0 \text{ 时等价}) \end{matrix}$   $\downarrow$  Defined in a neighborhood of  $\mathcal{M}_0$

**Reduced Problem:** slow times  $\varepsilon = 0$ . **Layer Problem:** fast time  $\varepsilon = 0$ . **Critical Manifold:**  $\mathcal{M}_0 := \{(u, v) : f(u, v, 0) = 0\}$  ps: Reduced problem defined only on  $\mathcal{M}_0$ , Layer problem defined off  $\mathcal{M}_0$

**Normal Hyperbolicity:**  $\mathcal{M}_0$  is normally hyperbolic at  $(u, v) \in \mathcal{M}_0$  if:  $\lambda_i$  (特征值) of Jacobian  $J|_{(u, v, 0)}$  are  $\frac{\partial f}{\partial u}|_{(u, v, 0)}$  are  $\Re(\lambda_i) \neq 0$ .  $\rightarrow$  **(Un-)Stable:**  $\Re(\lambda_i) < 0 (> 0) \Leftrightarrow \frac{\partial f}{\partial u} < 0 (> 0)$  (1-dim)

**Outer Solution:** sol for slow time at  $\varepsilon = 0$ . (reduced problem) **Inner Solution:** sol for fast time at  $\varepsilon = 0$ . (layer problem)  $\uparrow$  attracting(stable) / repelling(unstable)

**Describe Reduced|Layer Problem:** <sup>1</sup> 如果能解具体表达式就解; 不好解情况下给出  $u', v'$  仅关于  $u, v$  的表达式; <sup>2</sup> 如果可以画一个图; <sup>3</sup> 分析 attracting/repelling (stable/unstable); <sup>4</sup> 分析  $\tau, t \rightarrow \infty$  时的行为

**Fenichel's Theorem:** If  $\mathcal{M}_0 \subset \{f(u, v, 0) = 0\}$  is compact (closed & bounded) and normally hyperbolic. Suppose  $f, g$  are smooth. Then for  $\varepsilon > 0$  (sufficiently small),  $\exists$  **Manifold**  $\mathcal{M}_\varepsilon$  s.t.  $\mathcal{O}(\varepsilon)$  close and diffeomorphic to  $\mathcal{M}_0$ . (i.e.  $\mathcal{M}_\varepsilon$  is locally invariant under the flow of the full system (上面左边的 system)). ps:  $\mathcal{M}_\varepsilon$  called **slow manifold**.

**Find  $\mathcal{M}_\varepsilon$ :** If  $\mathcal{M}_0 = \{(u, v) : u = p_0(v)\}$ , then  $\mathcal{M}_\varepsilon = \{(u, v) : u = p_\varepsilon(v)\}$  可通过算  $u, v$  的 exact sol, 寻找能使含  $\varepsilon$  项消失但保留  $\tau$  的解 (or: 考虑快变量  $t \rightarrow \infty$ , 对比  $\mathcal{M}_0$  得到  $\mathcal{M}_\varepsilon$   $\downarrow$  **Infective Period:**  $= \frac{1}{\alpha}$ )

**SIR:** S: # people who are susceptible. (易感人群) I: # people who are infected. (感染者) R: # people who have been infected and recovered. (no longer susceptible). (康复者) **Total Pop:**  $N = S + I + R$

**Model:**  $S' = -\beta SI, I' = \beta SI - \alpha I, R' = \alpha I, S, I \geq 0$ . **Assumptions:** <sup>1</sup> Mass Action incidence:  $\beta N$  transmit infection/unit time; <sup>2</sup>  $\alpha I$  recover/unit time; <sup>3</sup> No im/emigration, births, deaths. <sup>4</sup> No deaths from disease.  $\rightarrow$  新感人数  $= \beta SI$  rate.

**临界值:** if  $S > \frac{\alpha}{\beta}, I \uparrow$  疾病爆发; If  $S < \frac{\alpha}{\beta}, I \downarrow$  疾病消退.  $\rightarrow$  **Basic Reproduction Number:** (A threshold quantity)  $R_0 = \frac{\beta S_0}{\alpha}$  or  $\frac{\beta N}{\alpha}$ , <sup>1</sup>  $R_0 > 1$  epidemic occurs; <sup>2</sup>  $R_0 < 1$  disease dies out.

$\cdot I_\infty = \lim_{t \rightarrow \infty} I(t) = 0; S_\infty = (S + I)_\infty > 0; N = S_0 + I_0$  **Final Size Relation:**  $\ln(\frac{S_0}{S_\infty}) = \frac{\beta}{\alpha} [N - S_\infty] = R_0 [1 - \frac{S_\infty}{N}]$  As  $(S + I)' = -\alpha I \rightarrow N - S_\infty = \alpha \int_0^\infty I dt$ ; As  $\frac{S'}{S} = -\beta I \rightarrow \ln(\frac{S_0}{S_\infty}) = \beta \int_0^\infty I dt$

**Attack Rate/Ratio:**  $N - S_\infty$  (# people ever infected | 总感染人数 | Final size of epidemic) **Orbits Sol:**  $I(t) + S(t) - \frac{\alpha}{\beta} \log(S(t)) = N - \frac{\alpha}{\beta} \log(S_0)$  **Peak:**  $I_{\max}$  when  $S = \frac{\alpha}{\beta} \leftrightarrow I' = 0$  (代入前面得精确  $I_{\max}$ )

## 6 Appendix

$y' + p(x)y = q(x)$ : **Integrating Factor:**  $I(x) = e^{\int p(x)dx} \Rightarrow I(x)y = \int I(x)q(x)dx$  **Taylor Expansion:**  $f(x, r)$  at  $(x_0, r_0)$ :  $f(x, r) = f(x_0, r_0) + (x - x_0)f_x|_{(x_0, r_0)} + (r - r_0)f_r|_{(x_0, r_0)} + \frac{1}{2} \dots$

**Binomial Series:**  $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$  **Linerisation:**  $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$   $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \dots$   $\ln(a + x) = \ln(a) + \frac{x}{a} - \frac{x^2}{2a^2} + \frac{x^3}{3a^3} - \dots$   $y(a) - y(b) = \int_b^a y'(x)dx$

**Hyperbolic Functions:**  $\sinh x = \frac{e^x - e^{-x}}{2}$   $\cosh x = \frac{e^x + e^{-x}}{2}$   $\tanh x = \frac{\sinh x}{\cosh x}$   $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$   $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$   $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$